

# TECHNICAL WHITEPAPER: WATERPRISM

## Real-Time Multi-Hazard Water Risk Intelligence & Autonomous Infrastructure Control

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**EXECUTIVE ABSTRACT:** Modern water infrastructure management is severely hindered by domain fragmentation. Traditional systems evaluate flood, drought, and water-quality degradation in isolation using static rule-sets. This creates critical blind spots during compound hazard events. **WaterPrism** resolves this via a unified, local-first intelligence platform integrating **multi-source spatial sensor fusion** (UK EA Flood API, FMI WFS, ERA5 reanalysis), **concurrent predictive risk modelling** (Flood analogues, FAO-56 Penman-Monteith ET0, WFD ecological proxies), and an autonomous **PPO reinforcement learning agent** (trained across 503,808 steps) controlling five simultaneous infrastructure actions (sluice gates, pumps, pre-drain rates, barriers, alert levels). A dedicated geopolitical layer tracks six global water choke points with automated UN/JRC GDACS feeds.

### 1. The Multi-Hazard Water Crisis & 4-Layer Architecture

Conventional platforms fail during compound events due to domain silos. WaterPrism unifies spatial sensor fusion, multi-hazard risk synthesis, and autonomous closed-loop actuator control across 4 coupled layers:

### 2. Spatial Sensor Fusion & Predictive Multi-Risk Modeling

**Inverse Distance Weighting (IDW):** Aggregates 9 telemetry nodes with  $p=2.0$  distance decay, standardizing precipitation, river stage, soil moisture, turbidity, and evapotranspiration into a continuous 12-dimensional state vector.

- **Flood Risk Engine:** Executes Euclidean analogue pattern matching over 18 historical storm profiles (2005–2023), generating 48–168 hour predictive hydrograph lead-times when  $D \leq 0.25$ .
- **Drought Stress Engine:** Composite score weighting 30-day precipitation deficit (40%), EA percentile rank (35%), and FAO-56 Penman-Monteith ET0 stress (25%), mapped to WMO categories.
- **Water Quality Engine:** Fuses turbidity and chlorophyll-a against EU Water Framework Directive ecological thresholds to predict runoff pollution surges before intake contamination.

### 3. Autonomous PPO Reinforcement Learning Benchmark

To coordinate mechanical assets without human latency, WaterPrism deploys an Actor-Critic Proximal Policy Optimisation agent across 325 ERA5-augmented storm episodes:

| Parameter / Metric | Simulated Baseline | PPO Policy Result   | Operational Impact                        |
|--------------------|--------------------|---------------------|-------------------------------------------|
| Training Steps     | 503,808 steps      | Converged (325 eps) | Zero policy collapse across extremes      |
| Cumulative Reward  | Mean: 27.265       | Optimal Control     | Zero flood breaches & minimal pump energy |
| Decision Latency   | < 1.2 ms / step    | Edge CPU / Vulkan   | Immediate local execution with 0 API cost |
| Actuator Degrees   | 5 Simultaneous     | Coordinated Action  | Sluice Gates, Pumps, Barriers, Alerts     |

### 4. Geopolitical Water Security & Global Threat Matrix

Monitors 6 strategic international water infrastructure nodes with real-time automated UN/JRC GDACS ingestion:

| Node ID | Region & Infrastructure Target | Threat Level | Intelligence Source |
|---------|--------------------------------|--------------|---------------------|
|---------|--------------------------------|--------------|---------------------|

|       |                                           |          |                              |
|-------|-------------------------------------------|----------|------------------------------|
| WST-1 | Gaza Catchment & Desalination Plants      | CRITICAL | UN OCHA / UNEP Operational   |
| WST-2 | Red Sea & Bab-el-Mandeb Chemical Corridor | CRITICAL | UKMTO / IMO Maritime Feeds   |
| WST-3 | Yemen Coastal RO Desalination Facilities  | CRITICAL | UN OCHA / WFP Advisories     |
| WST-4 | Strait of Hormuz GCC Sea Water Intakes    | HIGH     | Regional Marine Advisories   |
| WST-5 | Ukraine Kakhovka Dam Downstream Zone      | CRITICAL | EU Copernicus / GDACS Alerts |
| WST-6 | Gulf of Oman Commercial Tanker Feeds      | HIGH     | IMO Maritime Intelligence    |

### 5. Strategic Value Proposition & Enterprise Contact

**Commercialization & Catchment Deployment:** WaterPrism is engineered for immediate turnkey integration with municipal water authorities, civil protection agencies, and sovereign infrastructure operators. Powered by Sampo AI OS for deterministic zero-cloud execution.

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